

Implementation of bots resulted in 47 per cent return on investment (ROI). The scalable and responsive processes thus used provided increased operational agility and boosted revenue of the company. Human effort hours per year saved after automation came down to 3441. There was 23 per cent reduction in average time spent in performing background checks.

Optimisation also led to more than 6000 hours of manual effort and coverage of more than 64 tasks. Accuracy is now hundred per cent as there are

no human errors and biases. This has been a boon to the company as it witnessed an effective overall growth in its performance and resource utilisation. Going by the result, the number of bots will only increase with time.

Today, Mindtree is implementing automation technologies in almost all its verticals and reaping huge benefits. Many other companies also are doing the same to progress in today's dynamic business environment. Yet, only a few major companies agree to have extensively used automation across all their divisions.

Organisations need to realise that redesigning jobs is essential to integrate traditional ones with new-age technologies. The focus must be on replacing manual processes that were applied to perform redundant tasks by automation. This promises good ROI at low risk.

It is also necessary for them to understand that automation is complementary to the workforce and not a threat. Automation will create jobs for humans and will allow them to apply their thinking and decision-making skills to value-added jobs.

—Ayushee Sharma

2 SMART WATER MANAGEMENT

Water-related data includes quality, pressure, level, dissolved oxygen and more. Data management is essential to keep track of this large amount of sensor and meter data collected in real time

Water scarcity is caused mainly by non-revenue water, which is the water lost before it is available for consumption. Physical loss due to leakage from such reasons as corrosion and material degradation remain the worst offenders. Others include theft through water tapping, supply for putting out fires, etc.

For dealing with failure of pipelines, direct methods include techniques like cutting a section in the pipe. These are expensive and require manual labour. Although, indirect ones involving sensors and radar do not require exposing the pipe, these are still costly and short-ranged.

The long-term solution to consuming water sustainably is smart water management. A smart water grid system covers everything ranging from storage to consumption. Smart water solutions incorporate both hardware and software, such as smart water meters, wireless sensors (for communication) and analytics, for optimisation of the water cycle.

Real-time intelligent monitoring at the operators' end aids in mitigating possible risks like contamination of water. At the consumers' end, real-



Automated meter infrastructure and smart water metering
(Credit: www.allianceforwaterefficiency.org)

time feedback enables cost-cutting and water conservation. In this article, we focus on smart water meters.

According to a report by Research-andMarkets.com, the global smart meters market was at US\$ 8.6 billion in 2018 and is projected to grow at a CAGR of over seven per cent to reach US\$ 12 billion by 2024 owing to the growth of smart cities.

It can be observed that meter readings taken manually are often inaccurate. Also, in harsh weather conditions, measurements cannot be done and so estimations have to be made. Bills are generated from these readings, which are usually obtained once or twice a month.

Smart meters, on the other hand, record water consumption at a frequent pace and provide information reports daily for both customers and

utilities. Automatic billing is generated based on usage, doing away with the practice of the same billing for every apartment. These provide security from tampering attempts and accuracy better than manual meters.

For monitoring at remote locations, wireless communication is used to establish a connection with local area network (LAN) or wide area network (WAN). Meters can be powered by batteries, and have a life span of about twenty years.

Automatic meter reading systems ensure transparency at customers' end. Customers know about their consumption continuously and are notified of any irregularity that might occur due to problems in the systems. Automatic meter reading is gradually getting replaced by advanced metering infrastructure since data is accessible in real time. Unlike automatic meter reading, advanced metering infrastructure transmits data bidirectionally, to and from the customer and the supply chain.

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